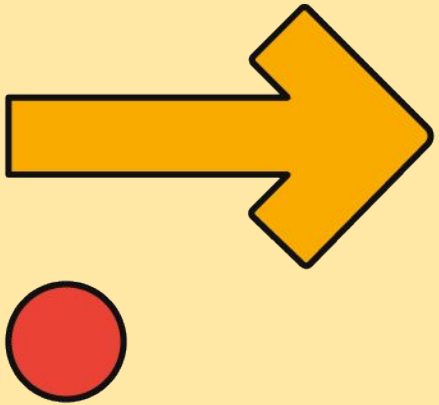
DevFest 2026



Intro to Cloud TPU

Gemma 4 on a v5e chip, managed over MCP

William McLean (xbill) · GDE Cloud + AI

About Me

DevFest

William McLean (xbill)

- AWS Community Builder 2026, AI
- GDE (Google Developer Expert), Cloud and AI/ML
- CTO and co-founder, Glitnir Ticketing
- 2026 goal: multi-cloud agents, tools and models

Where to Find Me

LinkedIn linkedin.com/in/xbill
Dev.to dev.to/xbill
Medium medium.com/@xbill999
GitHub github.com/xbill9
Hugging Face huggingface.co/xbill9

Agenda

DevFest

01 Ten years of TPUs

02 The chips, and what they compute natively

03 Getting a chip: control planes, quota and price

04 Live demo: Gemma 4 on a v5e-1 over MCP

05 So, which one?

DevFest

Part 01

Ten Years of TPUs



Google Developer Groups

The First TPU Was Built for Inference

DevFest

Year	Chip	What changed
2015	TPU v1	Inference only, 8-bit, 92 TOPS; announced 2016
2017	TPU v2	bf16 training, 180 TFLOPs per device, first pods
2019	TPU v3 pods	Liquid cooled, 1,024 chips, public beta
2021	TPU v4	Optical circuit switches, 4,096 chips
2023	v5e and v5p	Cost-optimized v5e; v5p with 95 GiB HBM
2024	v6e Trillium	4.7x v5e peak compute
2025	TPU7x Ironwood	fp8, 192 GiB HBM; “first TPU for the age of inference”; GA March 2026
2026	TPU 8t and 8i	Announced April 2026; 8i is the inference part

Blue rows: the inference-first chips. Sources: Google Cloud and Google blogs 2016 to 2026; Jouppi et al., arXiv 1704.04760.

vLLM Reaches the TPU Through tpu-inference

DevFest

Client
OpenAI API

vLLM
batching, KV paging

tpu-inference
JAX model code

XLA
one program per shape

TPU VM
1 to 8 chips per host

- tpu-inference v0.31.0 shipped 2026-10-08; v5e, v6e and v7x recommended, v5p experimental
- Gemma 4 E2B, E4B, 26B-A4B and 31B pass its release matrix
- XLA compile is 81 to 91% of engine start for E2B on v6e and v5e
- Pin an image digest: the nightly tag broke two 4-bit builds on 2026-10-02

Sources: github.com/vllm-project/tpu-inference; gemma4-dev tpu-jax-v5e1-2b and tpu-vllm-v5e1-2b reports.

DevFest

Part 02



The Chips

Google Developer Groups

TPU7x: About 13x v5e's Memory at 10x the Price

DevFest

Chip	HBM	Peak bf16	Native low precision	Max pod	On-demand
v5e	16 GB	197 TFLOPs	int8, 393 TOPs	256	$1.20/hr
v6e Trillium	32 GB	918 TFLOPs	int8, 1,836 TOPs	256	$2.70/hr
v5p	95 GiB	459 TFLOPs	int8; fp8 at bf16 rate	8,960	$4.20/hr
TPU7x Ironwood	192 GiB	2,307 TFLOPs	fp8, 4,614 TFLOPs	9,216	$12.00/hr

- Single-host shapes used here: v5litepod-1 and -4, v6e-1, -4 and -8
- v6e on-demand varies by region: $2.70 us-east, $2.97 europe-west4

Sources: docs.cloud.google.com/tpu/docs and cloud.google.com/tpu/pricing, read 2026-10-12; regional rates from the Billing Catalog. Blue row: today's demo chip.

One-Stream Decode Follows Memory Bandwidth

DevFest

Chip	Memory	Bandwidth	E2B bf16 tok/s, 1 stream	tok/s per TB/s
TPU v5e	16 GB	859 GB/s[1]	120	140
TPU v6e	32 GB	1,638 GB/s	199	121
TPU v5p	95 GiB	2,765 GB/s	not measured	—
TPU7x	192 GiB	7,380 GB/s	not measured	—
NVIDIA L4	24 GB	300 GB/s	46	153

- Each token reads every weight once, so bandwidth sets the ceiling for one request
- The three measured chips land at 121 to 153 tok/s per TB/s of bandwidth

[1] Google quotes v5e as 800 GiBps. Different vLLM builds per platform. Sources: gemma4-dev HARDWARE.md, MI300X dollar-hour article; Google TPU docs.

Only int8, and fp8 on TPU7x, Buy Compute

DevFest

Format	v5e	v6e	v5p	TPU7x
bf16	native	native	native	native
int8	2x bf16	2x bf16	native	native
fp8	storage	storage	bf16 rate	2x bf16
int4	storage	storage	storage	storage

- 2x bf16: a native matmul path at double the bf16 peak
- storage: saves HBM and bandwidth, then widens to bf16 before every matmul
- fp8 KV cache on v6e logs fp8 and allocates bf16

Sources: gemma4-dev HARDWARE.md, QUANTIZATION.md; Google TPU7x docs (not measured here).

“v5e-1” Works in Prose, Never in a gcloud Flag

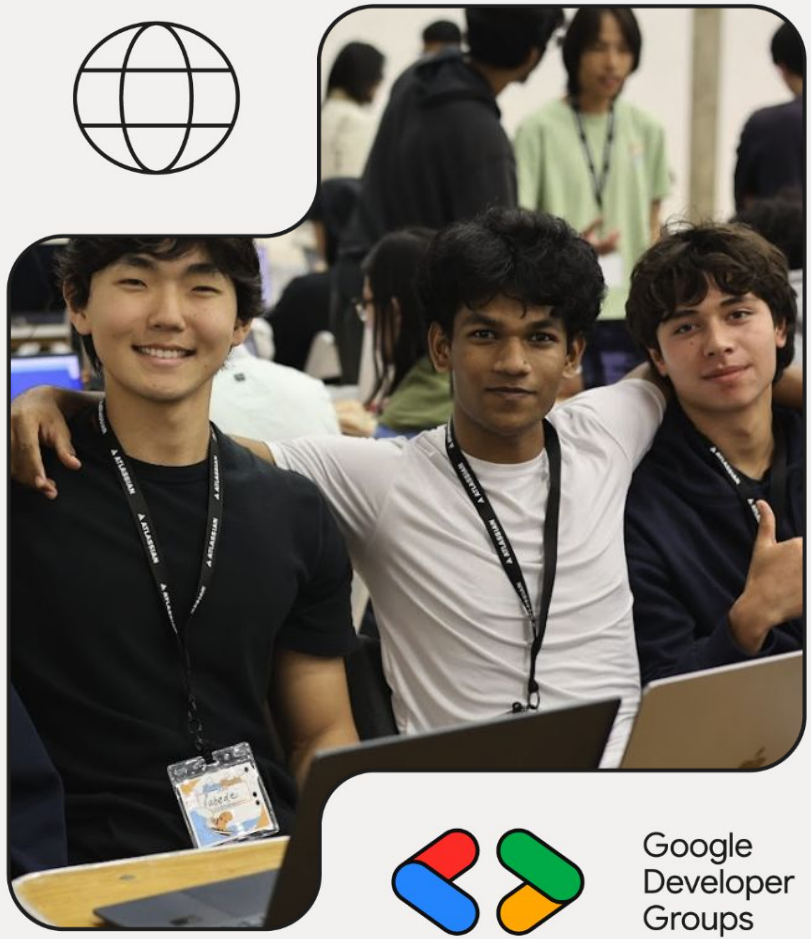
DevFest

What you mean	What gcloud wants	Where it goes
v5e, one chip	`v5litepod-1`	--accelerator-type (Cloud TPU API)
v5e runtime	`v2-alpha-tpuv5-lite`	--runtime-version (Cloud TPU API)
v5e machine	`ct5lp-hightpu-1t`	listed in the Compute Engine catalog; create is refused
v6e, one chip	`v6e-1` / `ct6e-standard-1t`	Cloud TPU API / Compute Engine
v6e runtime	`v2-alpha-tpuv6e`	--runtime-version (Cloud TPU API)

DevFest

Part 03

Getting a Chip



Google Developer Groups

Two Control Planes

DevFest

Cloud TPU API

- Queued resources produce a node named <id>-node
- gcloud alpha compute tpus queued-resources create
- Bug fixes and security updates only
- The only path for v5e outside GKE

Compute Engine

- The instance is the node: one object to manage
- gcloud compute instances create --machine-type=ct6e-standard-1t
- Supports TPU7x, v6e and v5p
- TPU7x (Ironwood) onward: Compute Engine or GKE only

“The Cloud TPU API is no longer under active development ... New hardware generations, starting with TPU7x (Ironwood), are supported only through Compute Engine or Google Kubernetes Engine (GKE).”

Quote: Cloud TPU resources in Compute Engine, Google Cloud docs.

Moving a v6e-1 Across: the Flag Mapping

DevFest

Cloud TPU API	Compute Engine
`--accelerator-type=v6e-1`	`--machine-type=ct6e-standard-1t`
`--runtime-version=v2-alpha-tpuv6e`	`--image-family=ubuntu-accel-… + --image-project`
`--valid-until-duration`	`--request-valid-for-duration`
`--provisioning-model=flex-start`	`--provisioning-model=FLEX_START`
`queued resource produces <id>-node`	`the instance is the node`
`gcloud compute tpus tpu-vm list`	`gcloud compute instances list`
`gcloud compute tpus tpu-vm ssh`	`gcloud compute ssh`

Same chip, same checkpoint, same serving flags: the KV cache came out at 1,151,744 tokens on both control planes.

Source: The unofficial TPU migration guide, dev.to/gde, 2026-08-11.

v5e Has No Compute Engine Create Path

DevFest

```
$ gcloud compute machine-types list --filter="name~ct5lp"
ct5lp-hightpu-1t  ct5lp-hightpu-4t  ct5lp-hightpu-8t   (26 zones)

$ gcloud compute instances create … --machine-type=ct5lp-hightpu-1t
ERROR: This user agent is not allowed to use the machine type [ct5lp-hightpu-1t].
```

- The catalog lists the machine type; GKE uses those names for its v5e node pools
- Google's TPU machines page names TPU7x, v6e and v5p for Compute Engine, and no v5e
- The v5e page: supported using Google Kubernetes Engine and the Cloud TPU API
- To test any chip, try a create in a zone where your quota is zero: a refusal costs nothing
- Today's demo runs on a Cloud TPU API queued resource

Four Ways to Ask for a Chip

DevFest

Model	Capacity	Max run	Ends	Quota spent	v6e, europe-west4
`STANDARD`	now, if available	unlimited	when you delete it	standard	$2.97 / chip-hr
`SPOT`	now, if available	unlimited	preempted at any time	preemptible → standard	$1.78 / chip-hr
`FLEX_START`	queues up to 2h	10 min – 7 days	at --max-run-duration	preemptible → standard	$1.35 / chip-hr
`RESERVATION_BOUND`	reserved ahead, if approved	up to 90 days	when the reservation ends	with the reservation	no list rate

- Flex-start is the cheapest of the three priced models on v6e, under half of on-demand
- On v5e in us-west4: flex-start $0.60, spot $0.607 per chip-hr
- The demo uses flex-start: it queues for capacity and deletes itself after 4 hours

Behaviour: Compute Engine provisioning models docs. Prices: Cloud Billing Catalog, read 2026-08-11.

Flex-Start Halves the On-Demand Rate

DevFest

$ per chip-hour	On-demand	Flex-start	Calendar	1-yr CUD	3-yr CUD	Spot read
v5e	$1.20	$0.60	$0.84	$0.84	$0.54	$0.58
v6e Trillium	$2.70	$1.35	$1.89	$1.89	$1.22	$1.40
v5p	$4.20	$2.10	$2.94	$2.94	$1.89	—
TPU7x Ironwood	$12.00	$6.00	$8.40	$8.40	$5.40	—

- Flex-start can undercut spot: v6e in us-east5 is $1.35 flex-start against $1.40 spot
- Flex-start runs 10 min to 7 days and bills until deleted
- TPU7x spot, flex-start and calendar are behind an allowlist

List rates: cloud.google.com/tpu/pricing, read 2026-10-12. Spot: Billing Catalog reads, Aug 2026.

Quota Does Not Mean Availability

DevFest

1. Hardware
in the zone

→

2. Provisioning
model offered

→

3. Capacity
right now

- 37 zones report v6e quota; 18 offer v6e-1 hardware, and Google's docs list 8
- 3 of 5 zones holding 1,536 chips of quota returned stockout
- us-central1-b has hardware, quota and a flex-start price, and rejects flex-start
- Probe with spot: it fails fast with “reason: stockout”; flex-start with no quota queues as PENDING, like a stockout
- v5e-1 flex-start: accepted in us-west4-a, rejected in europe-west4-a and -b

Source: gemma4-dev tpu_zones_status.md (2026-08-10), TPU.md, request-quota.sh; tpu-vllm-v5e1-2b CLAUDE.md (2026-08-04).

The Control Plane Fails Quietly

DevFest

Cloud TPU API is frozen
Bug fixes only. TPU7x runs on Compute Engine or GKE.

v5e has no Compute Engine path
Listed in 26 zones; create refused.

Two separate quota pools
us-east5: 512 v6e chips via the TPU API, 0 via Compute Engine.

RUNNING means booted
A GKE node is Ready about 10 min before the model serves.

Boot defaults break vLLM
10 GB disk, no Docker on PATH; a missing scope spins 30 min.

Nothing stops billing
GKE pools and old queued resources bill until deleted.

Sources: Google TPU docs; gemma4-dev TPU.md, request-quota.sh, gke-vllm-v6e1-2b.

DevFest

Part 04

Live Demo



Google Developer Groups

What the Agent Drives

DevFest



- The client talks to server.py over stdio; its tools wrap gcloud to provision, find the node's IP and start the container
- Queries go to vLLM's OpenAI-compatible /v1/chat/completions
- analyze_cloud_logging sends the TPU's own error logs to the model it hosts
- Project, zone, model and provisioning model come from one file, tpu.env

Demo Script

DevFest

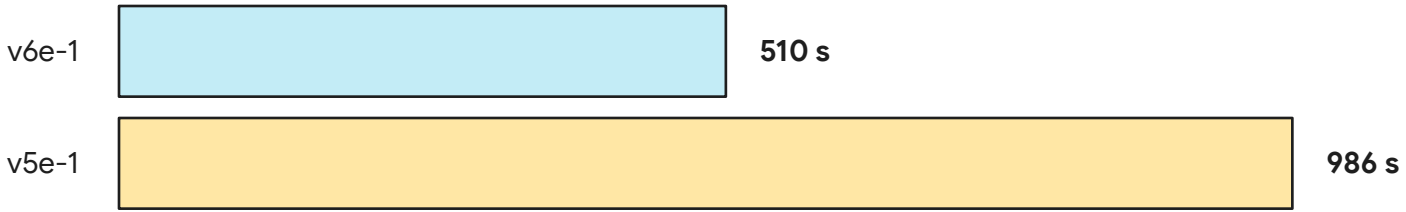
#	Ask the agent	Tool it calls
1	What's running in the zone?	`get_system_status`
2	Which zones have v5e quota?	`get_zones_with_available_quota`
3	Create the v5e-1, flex-start	`create_tpu_queued_resource`
4	Is it up yet? Show the container log	`describe_queued_resource, get_vllm_docker_logs`
5	Is the model healthy?	`verify_model_health`
6	Explain a TPU in two sentences, with speed	`query_queued_gemma4_with_stats`
7	Any TPU errors in the last hour?	`analyze_cloud_logging`
8	Tear it down	`destroy_queued_resource`

In a shared zone, provision with create_tpu_queued_resource: manage_queued_resource deletes every other queued resource in the zone.

XLA Compile Is Most of a Cold Start: Cache It

DevFest

Seconds from container start to healthy, Gemma 4 E2B



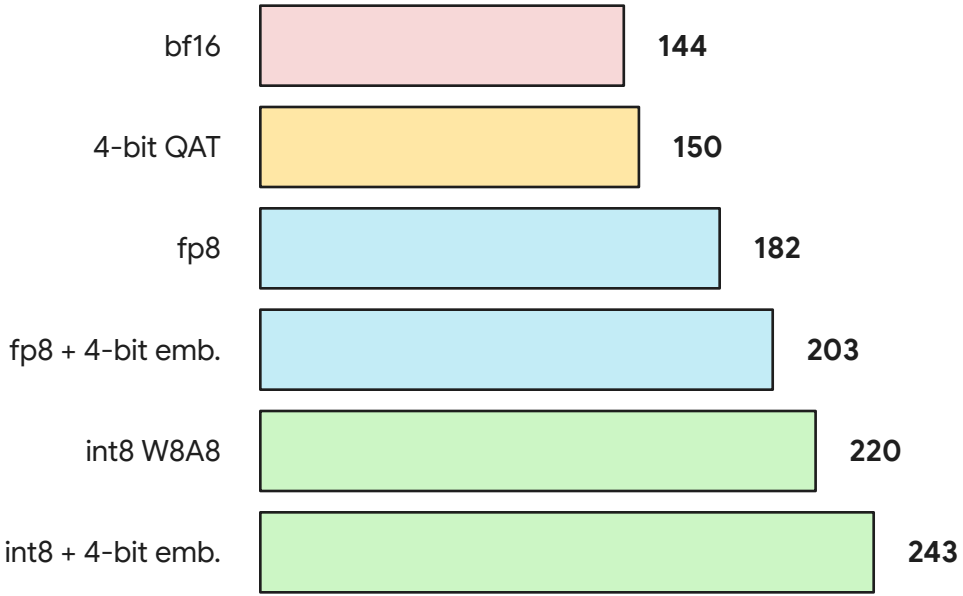
- XLA compile is 81 to 91% of engine start
- A mounted compile cache cut a v5e restart 42%, 857 to 497 s
- 31B weights load in 5.6 min from in-region GCS; 60 to 90 min from Hugging Face
- GKE from nothing to first token: about 28 min
- Each v6e cold start costs $0.19 to $0.24 at flex-start rates

Source: gemma4-dev tpu-jax-v5e1-2b and tpu-vllm-v5e1-2b reports.

On v5e, int8 Serves E2B 1.7x Faster Than bf16

DevFest

Output tok/s, one stream, v5e-1 (E2B)



- Accuracy holds: 0.683 bf16, 0.677 int8 + 4-bit embeddings
- fp8 trails int8 at the same file size: 518 against 624 tok/s on 12B
- 12B int8 (11.31 GiB) is the largest model served on one 16 GB v5e chip

One stream, v5e-1. Source: gemma4-dev QUANTIZATION.md, measured 2026-09-29/30.

Measured on One v5e Chip

DevFest

Gemma 4 E2B-it, bf16, v5litepod-1	Result
GSM8K, all 1,319 problems, 2,048-token limit	0.910
GSM8K, 768-token limit	0.894
BFCL v3 simple, 400 single-tool calls	0.928
~1,000-token prompts, output tok/s at 1 / 16 requests	139 / 1,521
~3,600-token prompts, output tok/s at 1 / 16 requests	130 / 906
~3,600-token prompts, time to first token at 16 requests	1.10 s

One chip, 16 requests, ~1,000-token prompts: 1,521 output tok/s

Measured 2026-09-30. GSM8K and BFCL greedy. Each long prompt has its own prefix, so the prefix cache serves none of it. Source: tpu-vllm-v5e1-2b README.

So, Which One?

DevFest

Workload	Pick	Why
E2B or E4B, cost first	v5e-1, int8, one engine per chip	1.7x bf16 speed; lowest $/M at spot or flex-start
12B to 31B on one chip	v6e-1, 4-bit QAT W4A16	26B at 1,283 tok/s; 31B fits in 21.67 GiB
bf16 26B or 31B, long context	v6e-4, TP=4	about 2,000 tok/s
fp8 math, 192 GiB per chip	TPU7x	allowlisted; not measured here
Stock software, fast boot	L4, or Cloud Run L4	26B 4-bit loads unpatched; 334 s to healthy

Then: probe capacity with spot, ask for the smallest slice, pin the image digest, mount the compile cache.

Takeaways

DevFest

One v5e chip serves Gemma 4 E2B at about 1,500 output tok/s across 16 requests, for $0.60 an hour on flex-start

int8 serves E2B 1.7x faster than bf16 on v5e, with accuracy within a point

New TPU generations ship on Compute Engine and GKE; v5e stays on the Cloud TPU API

Probe capacity with a spot create; read quota by metric name

Health-check the port: RUNNING and ACTIVE only say the machine is up

An MCP server turns the whole lifecycle into prompts, including teardown

Links

DevFest

Rigs and results
github.com/xbill9/gemma4-dev

tpu-inference
github.com/vllm-project/tpu-inference

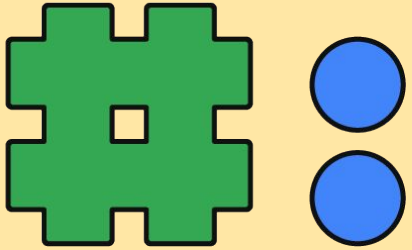
The unofficial TPU migration guide
dev.to/gde/the-unofficial-tpu-migration-guide-cloud-tpu-api-to-compute-engine-2co7

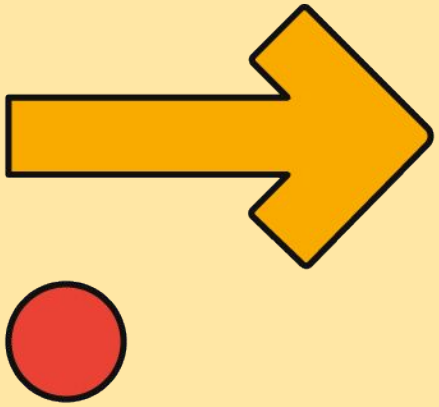
TPU docs and pricing
docs.cloud.google.com/tpu/docs
cloud.google.com/tpu/pricing

Where to Find Me

LinkedIn linkedin.com/in/xbill
Dev.to dev.to/xbill
Medium medium.com/@xbill999
GitHub github.com/xbill9
Hugging Face huggingface.co/xbill9







Questions?

github.com/xbill9/gemma4-dev

